

BALA SUBRAMANIAN L

Mechanical Design Engineer

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Portfolio: https://balasubramanian1609.github.io/My_Portfolio/ | Coimbatore, Tamil Nadu

PROFESSIONAL SUMMARY

Results-driven Mechanical Design Engineer with 2 years of R&D experience in pump design, hydraulic systems, CAD tools, SolidWorks, AutoCAD 2D drafting, GD&T, DFMEA, DFM/DFA and New Product Development (NPD). Skilled in tolerance stack-up analysis, Engineering change Management (ECM), BOM management, PLM tools (ENOVIA/SAP). Delivering reliable mechanical assemblies for waste water pump applications.

TECHNICAL SKILLS

Engineering Tools: AutoCAD, SolidWorks, MS Office, PLM (ENOVIA), SAP

Core Engineering: Mechanical Design, BOM preparation, Root Cause analysis (RCA), Hydraulic Design & Performance optimization, Failure Analysis & Reliability Improvement, Material selection.

Professional Skills: 3D Modeling, 2D Drafting, GD&T, DFMEA, DFM, DFA, NPSH, Specific speed, Tolerance Stack-up Analysis, Prototype Testing & Validation, Pump Performance curve & H-Q curve, Engineering Change Management (ECM), New product Development (NPD).

WORK EXPERIENCE

Design Engineer — CRI Pumps Private Limited

April 2026 – Present

Coimbatore, Tamil Nadu

- Engineered precision mechanical components of sewage and drainage pump including motor bodies, casings, Vortex impeller, Non-clog Impeller, bearing flanges, rotor shafts, and motor covers using SolidWorks, reducing design iteration cycles by 30% through improved first-pass accuracy.
- Developed complete mechanical assemblies for Sewage/Drainage pumps, Submersible Mixer pumps, Jet-type Aerators, and Submersible Agitators ensuring structural integrity and operational reliability.
- Conducted competitive analysis of 5+ market-leading pump brands to identify design gaps and implement targeted performance improvement opportunities.
- Executed design verification and modification of wastewater pump components through SolidWorks and engineering drawing reviews.
- Collaborated with cross-functional teams (Quality, Manufacturing) to resolve engineering issues, drive product validation, support prototype development, and achieve NPD milestones.

Associate Engineer — CRI Pumps Private Limited

September 2025 – April 2026

Coimbatore, Tamil Nadu

- Developed CAD models, technical drawings (GAD, CSD), and engineering documentation in SolidWorks and AutoCAD, ensuring design accuracy and drawing standards compliance.
- Generated and maintained item codes and EBOMs for 500+ components in ENOVIA PLM software, ensuring data accuracy and traceability across product lifecycle.
- Produced General Arrangement Drawings (GAD) and Technical Data Sheets (TDS) for the marketing team, enabling faster customer response and proposal turnaround time.
- Administered change requests (CR) and change orders (CO) through Engineering change Management (ECM) workflows, reducing field issues by 20%.

Graduate Engineer Trainee (GET) — CRI Pumps Private Limited

June 2024 – September 2025

Coimbatore, Tamil Nadu

- Mastered pump design fundamentals, hydraulic testing procedures, and R&D workflows through structured hands-on training and live project exposure across 3 product families.
- Collaborated with senior engineers to convert product concepts into detailed manufacturing drawings for machining and casting processes, applying Design for Manufacturability (DFM) principles.
- Maintained engineering documentation including test reports, BOM updates, and inspection records in SAP, supporting accurate configuration management and audit readiness.

KEY PROJECTS

Project 1: Optimization of Oil Chamber System to Improve Mechanical Seal Performance in Dry Pit Sewage Pumps

- Designed and implemented an innovative oil chamber system for vertical and horizontal dry pit sewage pumps improving mechanical seal lubrication and achieving a significant reduction in mechanical seal failure rates by 20%.
- Validated design effectiveness across 10 pump models (low to high kW range) under controlled R&D conditions, ensuring reliability and operational stability before production release.

Project 2: Upgrading Sewage pump Efficiency class from IE2 to IE3 Class

- Engineered and developed sewage pump models aligned with IE3 energy efficiency standards as part of R&D responsibilities, implementing design modifications that improved motor-pump system performance by 5% more than IE2.
- Optimized existing sewage pump designs to meet IE3 efficiency requirements, contributing to energy savings and regulatory compliance for the product range.

Project 3: Design & Development of Mixer pump Installation Stand for Type 11T & 12P & Auto Coupling Tank Configuration in Wastewater applications.

- Designed and developed structural installation stands for tank-mounted mixer pump configurations (11T and 12P, Auto coupling type), managing end-to-end NPD from concept through final design validation.
- Applied tolerance stack-up analysis and material selection principles to ensure structural integrity, long-term reliability in wastewater environments.

EDUCATION

B.E. – Mechanical Engineering 2021 – 2024

Coimbatore Institute of Technology, Coimbatore, Tamil Nadu | CGPA: 7.8/10

Diploma in Mechanical Engineering 2019 – 2021

S.Thangapazham Polytechnic College, Vasudevanallur, Tamil Nadu | Percentage: 99%

Higher Secondary Education – 12th 2019

H.N.U.C. Hr. Sec. School, Puliyangudi, Tamil Nadu | Percentage: 81.1%

SSLC 2017

H.N.U.C. Hr. Sec. School, Puliyangudi, Tamil Nadu | Percentage: 94.8%

CERTIFICATIONS & TRAINING

- AutoCAD 2D/3D Certification (level 1 & 2) – Autodesk
- SolidWorks – ARRIS CADD Centre
- SolidWorks Productivity Training – EGS Technical Training
- Inspection and Quality Control in Manufacturing - The Engineers Edge Institute of NDT
- Rapid Manufacturing – NPTEL
- Work System Design - NPTEL

ADDITIONAL INFORMATION

Languages: Tamil (Native) | English (Professional)

Location: Coimbatore, Tamilnadu — Open to Relocation